Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

Rewritten Chain of Thought

I may read only `prompt.md` and `OnShellBG.m`; everything else I must generate. The prompt is unusually generous in one respect: it tells me the answer is a *piecewise homogeneous polynomial*, that the kinematic space decomposes into *chambers*, and that a different homogeneous polynomial holds on each chamber. I decide right away to treat that not as flavour text but as the skeleton of the search, and I write it into my own words before touching the oracle: with $g = 1$ and $\sigma = (-1, -1, +1, \dots, +1)$ the external momenta are $k_i = \sigma_i \omega_i^2$, so the only place non-polynomial behaviour can enter is $|k_S| = \left|\sum_{i \in S} \sigma_i \omega_i^2\right|$ on *internal* Berends–Giele momenta. The sign of each internal sum is what cuts the chamber walls. I keep that observation in reserve and start probing.

First the environment: `wolframscript` and Python with exact rationals are both available. The demo block at the end of `OnShellBG.m` runs slow $n = 8$ cases I do not want, so I split the file at the test marker and load only the definition sections, then write a thin driver that builds the two-minus sign vector, calls `MakeKinematics`, and feeds `BGAmplitude`. The first data confirm the gross features the hint already half-promised: every A_n is *purely imaginary* and *homogeneous of degree* $2(n-2)$. So I set $A_n = -i\, P_n$ with P_n real and chase P_n. A representative symmetric point at $n = 5$ returns $A_5 = -2304\, i$; another gives $A_5 = -45824\, i/17$. Clean rational imaginary numbers, exactly as a polynomial answer should give.

The first real obstruction is $n = 4$. Every two-minus point the solver produces comes back `Indeterminate` — a $0 \cdot \infty$. I work out why by hand: for $n = 4$ the two conservation laws, after eliminating ω_1, collapse to

$$(\omega_2 + \omega_4)(\omega_2 + \omega_3) = 0,$$

so *every* on-shell point forces $\omega_2 + \omega_4 = 0$ or $\omega_2 + \omega_3 = 0$, and then an internal two-leg current hits zero frequency and zero momentum at once, blowing up a propagator behind a removable $0/0$. $n = 4$ is intrinsically degenerate: its kinematics is pinned to a chamber wall. I set it aside to recover later as a limit, and work with $n = 5, 6, 7$, which are generically clean.

Then I make the mistake the hint should have protected me from, and I want to be honest that I made it anyway. I check directly that A_5 is fully Bose-symmetric — invariant under swapping the two minus legs $1 \leftrightarrow 2$ and under permuting the plus legs $3, 4, 5$ — and confirm it is. From "symmetric + polynomial" I reason that P_5 must be a homogeneous degree-six polynomial in the elementary symmetric functions of the plus legs, and I try to fit it by exact rational linear algebra. It fails. I assume my sample points are clustered and rank-deficient, redo it with diverse exact sampling and a proper exact residual check, and it fails again — badly, with residual $\sim 3.6 \times 10^{10}$ even in exact arithmetic. That is a clean contradiction, not numerical noise: P_5 is provably *not* a polynomial in the plus-leg symmetric invariants, despite being symmetric.

The contradiction is exactly the hint cashing in. A globally symmetric function that is not a global polynomial in the symmetric invariants can only be a *piecewise* polynomial whose pieces are individually *not* symmetric — and the symmetry is restored by the way the chambers glue. Swapping $\omega_1 \leftrightarrow \omega_2$ does not stay inside a chamber; it crosses a wall. So the symmetric fit was doomed because it tried to bend one polynomial across walls it could not see. This is the moment the parked chamber idea moves to the centre of the work.

I stop trying to simplify the whole amplitude symbolically — the Vertex sums over $n!$ permutations of nested kernels blow up memory every time I ask Wolfram to `Simplify` them, and a univariate symbolic attempt drowns the same way. Instead I reconstruct A_5 along a one-parameter *line* by exact rational interpolation: evaluate at many exact points, recover the rational function, read off its denominator. The breakthrough is sharp. Along the line $\omega_3 = 4,\ \omega_4 = 9$, with $t = \omega_2$, I find that $A_5(t)\,(\omega_2+\omega_3+\omega_4)$ is a clean polynomial, and on the segment where ω_2 is the smallest scale it reduces to

$$A_5 \;=\; -16\,\omega_1\,\omega_2^5.$$

I check this against the scattered earlier data and it reproduces them: free frequencies $\{1, 4, 11\}$ give $\omega_1 = -49/4$, $\omega_2 = 1$ and $-16(-49/4)(1) = 196$; $\{2, 5, 11\}$ gives $59648/9$; $\{3, 7, 13\}$ gives $1434672/23$ — all matched. The earlier chamber-1 datum across n generalizes this to $A_n = -2^{\,n-1}\,\omega_1\omega_2^{2n-5}$, verified for $n = 5, 6, 7$. So in the chamber where ω_2 is the unique smallest-magnitude leg, the answer is a bare monomial.

Now I walk the line across the walls and reconstruct the polynomial on each chamber, and the corrections are too clean to miss. With $x = \omega_2^2$ and the plus squares ω_3^2, ω_4^2, the pieces are

$$x < \text{all}:\ \ -x^2, \qquad \omega_3^2 < x < \omega_4^2:\ \ -x^2 + (x-\omega_3^2)^2,$$

$$x > \omega_3^2+\omega_4^2:\ \ -x^2 + (x-\omega_3^2)^2 + (x-\omega_4^2)^2 - (x-\omega_3^2-\omega_4^2)^2 \;=\; -2\,\omega_3^2\omega_4^2,$$

each times the universal prefactor $16\,\omega_1\omega_2$. These are not three separate answers; they are the running inclusion–exclusion corrections of one object. Crossing the wall $\omega_2^2 = q_S$ (where $q_S = \sum_{j\in S}\omega_j^2$, S a subset of the plus legs) adds exactly $(-1)^{|S|+1}(\omega_2^2 - q_S)^{\,n-3}$. The walls are precisely the $|k_T|$ sign changes I had parked at the start. So the whole amplitude is one truncated-power sum,

$$A_n \;=\; 2^{\,n-1}\,\omega_1\omega_2 \sum_{\substack{S\subseteq\{3,\ldots,n\}\\ \omega_2^2 > q_S}} (-1)^{|S|+1}\,(\omega_2^2 - q_S)^{\,n-3}, \qquad q_S = \sum_{j\in S}\omega_j^2,$$

which is the bare monomial $-2^{\,n-1}\omega_1\omega_2^{2n-5}$ when only $S = \varnothing$ is active (the principal chamber) and grows term by term as ω_2^2 passes each subset-sum knot. Writing it with the truncated power $[\,u\,]_+ = \max(u, 0)$,

$$\boxed{A_n \;=\; -\,i\,\frac{2^{\,n-1}}{g^{\,n-3}}\,\omega_1\,\omega_2 \sum_{S\subseteq\{3,\ldots,n\}} (-1)^{|S|+1}\left[\,[\,\omega_2^2 - q_S\,]_+\right]^{\,n-3}}$$

a piecewise homogeneous polynomial of degree $2(n-2)$, exactly the shape the hint described. The chambers are the regions of constant active set $\mathcal{A} = \{S : \omega_2^2 > q_S\}$; on each, A_n is the single polynomial summing the active terms, and adjacent chambers differ by a term vanishing to order $n-3$ on the shared wall — a genuine spline.

Two consistency checks surprise me by working. First, although I wrote the threshold with ω_2^2, using ω_1^2 instead — or the smaller of the two — gives *identically zero residual* across all 21 points I test at $n = 5$, including points where ω_1 is the small one. I track this to a finite-difference identity: there are $n-2$ plus legs but the power is only $n-3$, so the full alternating sum $\sum_{S\subseteq\{3,\ldots,n\}}(-1)^{|S|}(x-q_S)^{n-3} \equiv 0$, an $(n-2)$-fold finite difference annihilating a degree-$(n-3)$ polynomial; together with $\omega_1^2+\omega_2^2 = q_{\{3..n\}}$ on shell this forces the two truncated sums to agree. I verify the identity symbolically for $n = 4, \ldots, 7$. Second, I had assumed the formula was g-independent and wrote that into the Python docstring; a quick check shows it is wrong — $A_n(g) = g^{-(n-3)}A_n(1)$ — so I restore the $g^{-(n-3)}$ factor (equivalently, $|k_j| = \omega_j^2/g$ makes the g's cancel inside the bracket). A small correction, but I had overclaimed and I fix it.

For $n = 4$ the spline has power $m = n-3 = 1$, so it is the continuous value $A_4 = i\,2^3\,\omega_1\omega_2^3$ at the degenerate locus the oracle could not evaluate. I confirm this honestly as a limit: taking the exact symbolic $\lim_{d\to 0}$ `BGAmplitude` along an energy-conserving deformation off the singular wall reproduces the formula at every test point — 24, 320, 1512 for the configurations I tried, including an $a > b$ relabeling. The removable $0/0$ comes out finite and matches.

Finally I verify, and here the only real friction is the machine, not the mathematics. `BGAmplitude` is slow at $n = 7$, so my first comprehensive script times out mid-run and a stress loop trips a paired-iterator `Do` syntax bug; I rewrite the verifier to find distinct chambers cheaply (the active set needs no oracle call) and invoke the expensive `BGAmplitude` only on one representative per chamber. The clean run lands: against the oracle in exact rational arithmetic the residual $A_{\text{formula}} - A_{\text{BG}}$ is *identically zero* for $n = 4$ (symbolic limit), $n = 5$ (4 chamber representatives plus a 76-point stress over 5 chambers, and explicit ω_1-smallest points), $n = 6$ (chambers of active-set size 1 through 8 plus a 38-point stress over 14 chambers), and $n = 7$ (12 distinct chambers whose polynomials carry up to 16 truncated-power terms). Every tested point matched with exactly zero residual, far inside the 10^{-10} tolerance the prompt asked for.

So I will state the scope plainly. The prompt gave the correct structural hint, and the run used it the way it was meant to be used: the chamber idea was parked at the first orientation, it was what turned the failed symmetric fit from a dead end into a diagnosis, and it named the walls ($|k_T|$ sign changes) before the data confirmed them. What the data still had to supply was everything quantitative — the principal-chamber monomial, the exact wall-crossing increment $(-1)^{|S|+1}(\omega_2^2 - q_S)^{n-3}$, the coefficient 2^{n-1}, the power $n - 3$, the g-scaling, and the inclusion–exclusion bookkeeping that assembles the pieces into one truncated-power sum. The result is a single all-kinematics closed form, valid in every chamber and at the degenerate $n = 4$ locus, verified against the oracle to exact equality.